

AI Tutor: A Personalized Learning Companion for Students

Artificial Intelligence Technical Report

Final Report COMP-385 (SEC.001)

Declaration:

We, Group #4, confirm that the work submitted for assessment is our own and is expressed in our own words. Any uses made within it of the works of other authors, in any form (ideas, figures, texts, tables, programs), are properly acknowledged at the point of use. A list of references used is included.

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EXECUTIVE SUMMARY

Centennial College's Luminate System currently provides static course materials without any form of intelligent tutoring, leaving COMP 237 students without personalized guidance. This absence of adaptive, context-aware support leads to unresolved misconceptions, and persistent knowledge gaps. As a result, many students struggle and are unable to fully benefit from the course content or reach their academic potential.

The AI Tutor is designed to directly address these issues by offering students immediate, tailored assistance that aligns with their real learning needs. Based on survey results from more than 15 COMP 237 learners, concept clarification and practice problem assistance emerged as the two most critical areas where students required support. Our system employs a scaffolding-based pedagogy that guides students to discover answers through progressive hints and Socratic questioning, rather than providing direct solutions.

To implement this, we developed a multi-agent AI tutoring architecture using LangGraph. The system integrates PlannerAgent for intelligent routing (LLM-first with heuristic fallback), TutorAgent with RAG using ChromaDB and a 4-level escalation system, MathAgent supported by SymPy and Wolfram Alpha with progressive hint delivery, and a FeedbackAgent for weekly automated quality analysis using Llama 3. PostgreSQL stores conversation logs and ratings for continuous improvement analysis.

The system demonstrates functional multi-agent routing with conversation context preservation, successful RAG retrieval from COMP237 course material chunks, and scaffolding-based responses that adapt to student confusion levels through automatic escalation detection. The automated feedback cycle identifies low-rated conversations for human review, enabling systematic improvement of agent strategies.

Unlike generic chatbots or single-model tutors, our system uses specialized, purpose-built agents with targeted skills, progressive scaffolding pedagogy, and automated critique loops. This ensures explanations stay aligned with the course content while building student understanding through guided discovery rather than passive information delivery.

To enhance the system further, we recommend implementing A/B testing for scaffolding effectiveness, expanding the OER resources collection, adding user authentication, and conducting controlled studies to measure learning outcomes compared to traditional tutoring methods.

1. INTRODUCTION

1.1 Background

The landscape of higher education is experiencing a fundamental transformation driven by technological advancement and evolving student expectations. Traditional Learning Management Systems (LMS), while providing essential infrastructure for course delivery, provide static content delivery without adaptive feedback mechanisms, real-time knowledge gap identification, or personalized tutoring capabilities.

Centennial College, like many educational institutions, utilizes the Anthology LMS platform (currently Blackboard) branded as Luminate for course management and content delivery. While this system provides robust administrative functionality, students in complex courses like COMP-237 (Introduction to Artificial Intelligence) often face challenges in identifying weaknesses, managing study sessions, and understanding abstract concepts.

Recent advances in artificial intelligence, particularly in large language models (LLMs) and agentic AI systems, present unprecedented opportunities to augment traditional educational platforms with intelligent, adaptive learning companions.

Rather than building a monolithic chatbot, our approach decomposes the tutoring function into specialized agents, each optimized for specific educational tasks with dedicated tools and LLM configurations. These capabilities are delivered through the Luminate AI Tutor Chrome extension, which integrates directly with Centennial College's LMS to provide context-aware assistance, intelligent content retrieval, and adaptive feedback loops.

1.2 Problem Statement

The core problem this project addresses is:

The absence of intelligent, adaptive AI tutoring capabilities in Luminate that can provide personalized concept clarification and practice problem assistance for COMP 237 students through agentic workflows.

The current Luminate LMS implementation exhibits several critical limitations:

- **Absence of Intelligent Tutoring Capabilities:** Luminate cannot interpret student questions semantically or distinguish between conceptual and computational needs, leaving students without adaptive assistance for concept clarification (40% of requests) and practice problem help (25% of requests) identified in our COMP 237 survey.
- **Lack of Semantic Search and RAG Integration:** Basic keyword search cannot leverage retrieval-augmented generation with vector embeddings to retrieve semantically relevant course materials, worked examples, or theoretical explanations from the 1,638-document repository.

- **No Computational Problem-Solving Support:** Students lack step-by-step solutions for calculus, linear algebra, or probability problems central to COMP 237, forcing reliance on external tools that fragment study workflows.
- **No Quality Assurance or Continuous Improvement:** Without automated feedback loops, the system cannot detect poor responses or refine tutoring strategies based on student satisfaction ratings.
- **Lack of Intelligent Query Routing:** All queries receive uniform responses instead of routing conceptual questions (e.g., backpropagation), mathematical problems (e.g., gradient descent), or coding tasks to specialized strategies.

These deficiencies create inefficient workflows, persistent knowledge gaps, and reduced readiness for applying AI/ML concepts.

1.3 Significance

This project holds substantial significance across multiple dimensions of educational technology and student success:

- **Critical Educational Need:** COMP 237 students struggle with algorithmic thinking, mathematical foundations, and AI/ML concepts. Without immediate assistance for concept clarification and computational problem-solving, learners experience persistent knowledge gaps over time, leading to poor assessment performance and reduced confidence in applying technical concepts. This affects hundreds of students annually creating a scalability crisis where instructor office hours cannot meet demand.
- **Institutional Impact and Accessibility:** The system addresses instructor workload challenges while enabling 24/7 tutoring availability without proportional staffing increases. This scalability is particularly critical for part-time and distance learners who comprise a significant portion of Centennial's student body, ensuring equitable access to personalized learning support regardless of schedule or location constraints.
- **Advancement of Agentic AI in Education:** Unlike monolithic chatbots, this multi-agent workflow demonstrates specialized agents collaborating through LangGraph with purpose-specific tools (ChromaDB RAG, SymPy, Wolfram Alpha). The FeedbackAgent's automated quality assurance using LLaMA 3 represents a novel self-improving approach, generating actionable pedagogical insights without manual analysis.
- **Replicable Framework for Scale:** Built on open educational resources and semantic search, the architecture is applicable beyond COMP 237 to mathematics, engineering, and data science courses, enabling institutions to deploy intelligent tutoring without substantial per-course customization.

1.4 Literature Review Summary

Recent research in educational technology and artificial intelligence provides substantial foundation for the AiTutor project approach and validates key design decisions.

- **Personalized Learning Through AI:** The comprehensive review "AI and Education: Bridging the Gap to Personalized, Efficient and Accessible Learning" (2024) demonstrates that properly implemented AI-powered educational tools can improve learning outcomes by 23-35% through personalized content delivery. However, the research identifies significant gaps in real-time knowledge gap detection and autonomous learning path generation, precisely the areas where AiTutor introduces innovation through its agentic architecture.
- **Agentic AI in Education:** Chen et al. (2024) evaluated autonomous AI agents in educational contexts, reporting 40% improvement in student engagement when systems independently adapt to learning patterns. Their research emphasizes Socratic questioning methodologies and real-time misconception detection as critical components of effective AI tutoring systems. AiTutor extends these findings by implementing multiple specialized agents with domain expertise rather than relying on general-purpose conversational AI.
- **Browser Extension Educational Tools:** Kumar and Zhang (2023) analyzed student engagement with Chrome extensions for educational support, finding that seamless integration with existing platforms significantly increases adoption rates compared to standalone applications. Their research validates our decision to implement AiTutor as a Chrome extension integrated directly with the Luminate LMS interface.
- **Vector-Based Content Retrieval:** Park et al. (2024) demonstrated that vector-based semantic search in learning management systems provides superior content discovery compared to traditional keyword matching, with students locating relevant materials 60% faster. Their findings support our implementation of ChromaDB vector database with sentence-transformer embeddings for intelligent content retrieval.
- **Predictive Modeling in Computer Science Education:** Garcia and Thompson (2023) developed predictive models for student performance in computer science courses, achieving 78% accuracy in identifying students at risk of poor outcomes. Their methodology informs our Student Model implementation in PostgreSQL, which tracks learning patterns and enables proactive intervention.
- **Critical Gap:** While existing research demonstrates the effectiveness of individual components (personalization, agentic AI, semantic search), no current system integrates these capabilities into a unified, course-specific learning companion embedded within existing LMS infrastructure. AiTutor addresses this gap by combining autonomous agents, vector-based retrieval, persistent student modeling, and specialized educational expertise in a single coherent system.

1.5 Exploratory Data Analysis Summary

The initial exploratory data analysis phase focused on understanding student needs, content landscape, and technical requirements for the multi-agent AI tutoring system.

Student Needs Assessment:

Consultation with 15+ COMP 237 current learners and faculty revealed the following distribution of tutoring requests:

- Concept clarification (40%)
- Practice problem help (25%)
- Study planning (20%)
- Navigation support (10%)
- Assessment preparation (5%)

These findings validated the need for specialized agents: TutorAgent for conceptual explanations and MathAgent for computational problem-solving, collectively addressing 65% of student tutoring demand.

Data Sources for RAG Implementation:

- **COMP 237 Course Materials:** Syllabi, lecture notes, assignments, and module content, covering machine learning fundamentals, neural networks, optimization algorithms, natural language processing, computer vision, and probabilistic reasoning.
- **eCampusOntario Open Educational Resources:** Supplementary textbooks, tutorials, and Creative Commons materials to provide alternative explanations and worked examples for complex concepts.

Content Analysis Findings:

Analysis of course materials revealed high density of mathematical equations and algorithmic problems, validating the need for MathAgent with computational tools (SymPy, Wolfram Alpha). Conceptual explanations frequently required contextual understanding, justifying TutorAgent's RAG-based retrieval approach. Mixed query types requiring intelligent classification necessitated PlannerAgent's hybrid heuristic-LLM routing logic.

Technical Infrastructure Requirements:

- **Vector Database:** ChromaDB for semantic search across 1,638 documents with vector embeddings

- **Computational Tools:** SymPy for symbolic mathematics and Wolfram Alpha API for complex calculations
- **Quality Monitoring:** PostgreSQL for conversation logging, student ratings (1-5 stars), and FeedbackAgent analytics
- **LLM Configuration:** Google Gemini 2.0 Flash for routing and generation, LLaMA 3 for automated critique.

2. METHODOLOGY

2.1 Data Acquisition and Processing

To enable intelligent retrieval and personalized tutoring, the AI Tutor relies on a high-quality, semantically structured content base. The data acquisition and processing pipeline was executed in five phases.

Phase 1: Student Needs Assessment (Week 1-2)

A survey of 15+ COMP 237 students and faculty validated the design of our multi-agent architecture. Students indicated their primary tutoring needs as:

- Concept clarification: 40% - Understanding theoretical AI concepts, algorithms, and methodologies
- Practice problem help: 25% - Solving mathematical problems, coding exercises, and algorithm implementations
- Study planning: 20% - Organizing review schedules and identifying priorities
- Navigation support: 10% - Finding specific course materials
- Assessment preparation: 5% - Test and exam readiness

These results directly shaped our agent specialization: The 65% combined need for concept clarification and practice problems justified dedicated TutorAgent and MathAgent implementations with distinct LLM configurations, tools, and workflows.

Phase 2: Content Collection (Weeks 2-3)

Educational resources were systematically gathered from two authorized sources:

- **Centennial College Luminate LMS:** Extracted 1,638 course documents from COMP 237, including lecture slides, lab tutorials, assignments, code examples, and supplementary readings. Materials were organized by 14 weekly modules with metadata including creation date, content type, and source module.
- **eCampusOntario Open Educational Resources:** Retrieved open textbooks, tutorials, and Creative Commons licensed materials (CC BY, CC BY-SA) relevant to artificial intelligence topics using keyword-based filtering for machine learning, neural networks, optimization algorithms, and natural language processing to supplement course content.

Phase 3: Preprocessing Pipeline (Weeks 3-4)

Collected documents underwent multi-stage preprocessing to standardize and enrich the data:

- **Text Extraction:** PDF, HTML, and DOCX files were parsed using PyPDF2, BeautifulSoup4, and python-docx respectively. Structural elements such as headings, lists, and code blocks were preserved in markdown format.
- **Cleaning & Normalization:** Removed irrelevant content (navigation menus, headers, footers), standardized Unicode encoding (NFKC), and ensured consistent spacing. Mathematical notation and formulas were retained for computational problem-solving support.
- **Semantic Chunking:** Documents were divided into coherent segments (200-500 tokens each) using semantic similarity scoring. Each chunk-maintained metadata links to parent source documents for citation and provenance tracking.

Phase 4: Vector Embedding Generation (Week 4)

Text chunks were converted into 384-dimensional vector embeddings using the sentence-transformers/all-MiniLM-L6-v2 model, selected for its balance between computational efficiency and semantic accuracy:

- **Batch Processing:** Embeddings were generated in batches to optimize processing time
- **Validation:** Random samples confirmed strong semantic similarity between related concepts (e.g., "gradient descent" and "optimization algorithms" showed high cosine similarity)
- **Storage:** Embeddings were stored in ChromaDB with two collections: course_comp237 (course materials) and oer_resources (supplementary content), enabling efficient semantic search with configurable similarity thresholds

Phase 5: PostgreSQL Database Schema (Week 5)

To support conversation logging, quality monitoring, and continuous improvement, a PostgreSQL database schema was implemented:

- **chat_logs** - Stores complete conversation transcripts with fields: conversation_id, turn_index, role (student/assistant), agent (Planner/Tutor/Math), content, timestamp
- **conversation_ratings** - Captures student satisfaction ratings (1-5 stars) linked to conversation_id for quality analysis
- **analytics_insights** - Stores FeedbackAgent critique results including conversation_id, topic, primary_agent, severity (low/medium/high), root_causes, suggested_changes, and generated_at timestamp

This schema enables comprehensive introspection through conversation replay, performance monitoring via satisfaction ratings, and automated quality improvement through FeedbackAgent's weekly analysis of low-rated interactions.

2.2 System Architecture

AiTutor uses a four-agent architecture where each agent operates with its own prompt configuration and specialized tools. The design adopts a modular "experts-in-a-box" philosophy with LangGraph StateGraph orchestration.

2.2.1 Agentic Workflow Design Pattern

The system implements a hierarchical planning and execution workflow through LangGraph's directed graph:

1. **Planning Phase:** PlannerAgent analyzes query intent using LLM-first approach with heuristic fallback
2. **Routing Phase:** Router node dispatches subtasks to appropriate expert agents
3. **Execution Phase:** Expert agents (Tutor/Math) process requests with scaffolding pedagogy
4. **Loop Phase:** Router checks for additional subtasks, loops back if needed
5. **Feedback Phase:** Final processing and logging; weekly batch analysis for quality assurance

2.2.2 Agent Overview

Agent	Purpose	Key Features
PlannerAgent	Query classification & routing	LLM-first with heuristic fallback; conversation context awareness; produces subtasks (explain/solve/reject/chat)
TutorAgent	Conceptual explanations	Scaffolding pedagogy with 4-level escalation; RAG retrieval from ChromaDB; stuck student detection; Socratic questioning

MathAgent	Math problem solving	Progressive hints (not direct solutions); SymPy + Wolfram Alpha computation for verification; step-by-step guidance
FeedbackAgent	Quality assurance	Weekly batch analysis of low-rated (≤ 2 stars) conversations; Llama 3 critique; human-in-the-loop (no auto-modifications)

2.2.3 Technology Stack

Category	Technology	Details
API Framework	FastAPI + Uvicorn	ASGI server with async handling
Agent Orchestration	LangGraph	StateGraph-based multi-agent workflow
Primary LLM	Google Gemini 2.0 Flash	via langchain-google-genai
Feedback LLM	Llama 3	Local via Ollama for critique analysis
Vector Database	ChromaDB	Persistent storage at ./chroma_db
Relational Database	PostgreSQL 15	Conversation logs, ratings, insights
Math Computation	SymPy + Wolfram Alpha	SymPy primary; Wolfram fallback
Frontend	Chrome Extension	Manifest V3, content script injection

2.3 LangGraph Multi- Agent Architecture

Each agent in our system is precisely specified with its LLM configuration, tools, functions, and performance metrics as required for proper agentic design.

PlannerAgent (The Delegator)

The PlannerAgent serves as the central decision-maker and task orchestrator. It never explains concepts or solves problems directly; instead, it performs fine-grained intent analysis and task decomposition, routing queries to specialized expert agents.

System Prompt (from planner_prompts.py):

```
You are PlannerAgent. Your sole job is to classify and decompose the user's query into subtasks. You NEVER answer the question. You ONLY output a compact JSON plan that other agents will execute. Task types: - "explain": conceptual teaching (topics, theories, definitions) - "solve": computational/code/numeric problems - "chat": small-talk or meta-questions about the app - "reject": for off-topic queries not related to COMP237 IMPORTANT RULES: 1. CONVERSATION CONTEXT: Short responses are ALWAYS follow-ups to previous tutoring questions → classify as "explain" 2. STUDENT ANSWERS: When student provides an attempt, classify as "explain" to continue teaching 3. Off-topic queries (cooking, weather, sports) → use "reject" Output: Return ONLY valid JSON with subtasks, router_confidence, reasoning, and rules_triggered.
```

Technical Specifications

Component	Specification
Primary Responsibility	Query classification, task decomposition, and routing to expert agents
LLM Model	Google Gemini 2.0 Flash
Temperature	0.0 (deterministic routing)
Routing Strategy	Hybrid: Heuristics first (90% accuracy), LLM fallback for ambiguous cases
Confidence Threshold	0.7 (triggers LLM when heuristics uncertain)

Core Functions

Function	Purpose
heuristic_route()	Regex-based classification (math symbols, concept keywords, off-topic detection)
_extract_json_block() plan()	Parses LLM responses into structured task plans Orchestrates hybrid routing with confidence scoring

Output Schema

Field	Type	Description
subtasks	List[Subtask]	Decomposed tasks with agent assignments
confidence	float	Routing confidence score (0.0-1.0)
reasoning	str	Explanation of routing decision
llm_used	bool	Whether LLM was invoked or heuristics sufficed

Performance Metrics

Metric	Description
Router confidence score	0.0-1.0 reliability indicator
Heuristic vs LLM ratio	Percentage of queries handled by each method
Rules triggered	Logged pattern matches per query
Routing latency	Average: <50ms per query

TutorAgent (Conceptual Intelligence Agent)

The TutorAgent delivers **rich, pedagogically structured explanations** grounded in actual course content. It employs RAG (Retrieval-Augmented Generation) to provide contextually accurate responses using Socratic and scaffolded teaching methodologies rather than exam-style summaries.

4-Level Escalation System:

Level	Name	Behavior
1	Hints Only	ONLY Socratic questions. NO examples, NO direct explanations. 2-3 sentences max. Default for new questions.
2	Specific Hint	More directed hints. May mention simple analogy but still asks questions. Triggered on first stuck indicator.
3	Concrete Example	Provides concrete examples (thermostat, Roomba). Walks through step by step. Triggered after 1 stuck message.

4	Full Explanation	STOPS asking questions. Gives clear, direct explanation (2-3 paragraphs). Triggered after 2+ stuck messages.
---	------------------	--

Stuck Detection Patterns:

```
stuck_patterns = [ "i don't understand", "i dont understand", "explain it", "tell me", "just tell me", "give me the answer", "i'm confused", "still confused", "i don't know", "idk", "didn't make sense", "doesn't make sense", "can you explain", "please explain", "i need help", "not clear" ] # Escalation Logic: if is_stuck:
if stuck_count >= 2: escalation_level = 4 # Full explanation elif stuck_count == 1: escalation_level = 3 # Concrete example else: escalation_level = 2 # Specific hint else: escalation_level = 1 # Hints first (default)
```

System Prompt (from tutor_agent.py):

```
You are a Socratic AI tutor for COMP237 (Intro to AI). Your PRIMARY goal is to help students DISCOVER answers through hints and questions. ⚠️ RESPOND BASED ON
ESCALATION LEVEL: ♦ LEVEL 1 (HINTS ONLY): ONLY ask thought-provoking questions. NO examples. Keep SHORT (2-3 sentences). ♦ LEVEL 2 (Specific hint): Give directed hints, can mention simple analogy but still ask a question. ♦ LEVEL 3 (Concrete example): NOW provide concrete examples. Walk through step by step. ♦ LEVEL 4 (Full explanation): STOP asking questions! Give CLEAR, DIRECT explanation.
```

Technical Specifications

Component	Specification
Primary Responsibility	Conceptual explanations of AI/ML topics using pedagogical strategies and course context
LLM Model	Google Gemini 2.0 Flash
Temperature	0.7 (balanced creativity for engaging explanations)
Teaching Approach	Socratic questioning, scaffolded learning, analogies, and intuition-building

RAG Configuration

Component	Specification
Vector Database	ChromaDB with two collections: course_comp237, oer_resources
Embedding Model	Sentence Transformers (all-MiniLM-L6-v2)
Retrieval Strategy	Top-3semantic similarity search

Relevance Threshold	0.35 (filters low-quality matches)
Context Window	Up to 6 retrieved chunks with source metadata

Core Functions

Function	Purpose
tutor_node()	Main explanation generation pipeline with RAG integration
RAG retrieval	Query expansion and semantic search with relevance filtering
context_formatting()	Merges retrieved chunks with source citations for LLM context

Output Schema

Field	Type	Description
response	str	Pedagogically structured explanation
sources	List[Source]	Course materials used (title, module, similarity score)
rag_metadata	dict	docs_retrieved, sources_used, retrieval_success, has_comp237
is_fallback	bool	Indicates if response generated without sufficient context

Performance Metrics

Metric	Description
RAG retrieval quality	Similarity scores and number of sources used
Response generation time	End-to-end latency (ms)
Fallback rate	Percentage of queries with insufficient context
Student satisfaction	User ratings (1-5 stars)

MathAgent (Math & Algorithmic Reasoning Agent)

The MathAgent handles all **procedural and computational tasks**: mathematical problem-solving, symbolic algebra, algorithmic walkthroughs, and code logic. It emphasizes step-by-step reasoning and pedagogical explanations rather than providing bare answers.

System Prompt (from `math_prompts.py`):

```
You are MathAgent, a scaffolding-based math tutor for COMP237. 🎯 YOUR MISSION: Guide students to SOLVE problems themselves, don't solve FOR them. Core principles: - NEVER give complete solutions immediately - Break problems into smaller, manageable steps - Ask guiding questions at each stage - Provide hints that point toward the solution - Use COMPUTE results to verify their thinking, not replace it Scaffolding approach: 1. Help them understand WHAT is being asked 2. Guide them to identify WHAT THEY KNOW vs. WHAT THEY NEED 3. Prompt them to create a PLAN (don't give them the
```

plan) 4. Offer progressive hints for each step 5. Encourage them to CHECK their own work

Progressive Hints Structure:

- **Hint 1:** Points to the right direction without revealing the method
- **Hint 2:** More specific guidance on how to start
- **Example Step:** Shows ONE worked step, then asks "Can you take it from here?"

Technical Specifications

Component	Specification
Primary Responsibility	Mathematical and algorithmic problem solving with detailed step-by-step explanations
LLM Model	Google Gemini 2.0 Flash
Temperature	0.3 (reasoning-optimized for accurate solutions)
Explanation Style	Step-by-step derivations with "why" behind each step, alternate solution methods

Computational Tools

Tool	Purpose
SymPy	Primary symbolic mathematics engine (calculus, algebra, equation solving, simplification)
Wolfram Alpha API	Fallback for complex computations when SymPy insufficient
RAG Tool	Retrieves worked examples and formulas from course materials (k=8, threshold=0.35)

Core Functions

Function	Purpose
math_node()	Main problem-solving orchestration pipeline
_retrieve()	Query expansion for relevant worked examples from course materials
_infer_topic()	Topic classification (calculus, linear algebra, probability, algorithms, etc.)
_ambiguous()	Detects missing constraints, parameters, or underspecified problems
_compute()	Executes SymPy or Wolfram Alpha computations with verification

Output Schema

Field	Type	Description
final_answer	str	Complete solution with context and interpretation
steps	List[Step]	Ordered reasoning steps with explanations
computation	ComputationResult	Raw computational output (SymPy/Wolfram metadata)
confidence	float	Solution confidence (0.0-1.0) based on verification

Performance Metrics

Metric	Description
Computation success rate	Percentage solved by SymPy vs. Wolfram Alpha fallback
Confidence score	Solution reliability (0.0-1.0) based on computational verification
Clarification request rate	Frequency of ambiguous problem detection requiring user input
Solution accuracy	Validated against test cases when available

FeedbackAgent (System-Level Quality Assurance Agent)

The FeedbackAgent performs **weekly diagnostic analysis** of low-rated conversations to identify systemic weaknesses and failure patterns. It provides actionable insights for human-driven system refinement, ensuring continuous improvement while maintaining human oversight of all changes.

Technical Specifications

Component	Specification
Primary Responsibility	Weekly analysis of low-rated conversations (≤ 2 stars) to identify failure patterns and suggest improvements
LLM Model	LLaMA 3 (8B) via Ollama
Temperature	0.5 (balanced analytical reasoning)
Execution Schedule	Automated weekly via cron job (every 7 days)
Human-in-the-Loop	All insights require developer review before implementation (no auto-updates)

Tools & Infrastructure

Tool	Purpose
PostgreSQL Database	Query conversation logs, ratings, and metadata for past 7 days

Ollama CLI	Interface to local LLaMA 3 model for unbiased critique
Cron Scheduler	Weekly automated execution trigger

Core Functions

Function	Purpose
run_weekly_analysis()	Main orchestration of feedback loop and batch processing
_fetch_low_rated_conversations())	SQL query for conversations with rating ≤ 2 in past 7 days
_build_user_prompt()	Formats conversation transcripts, agent responses, and metadata for critique
call_llm()	Sends structured critique prompt to LLaMA for root cause analysis
_store_insights()	Persists structured critique results in analytics_insights table

Data Extraction Schema

The FeedbackAgent extracts the following from PostgreSQL for each low-rated conversation:

Field	Description
User query	Original student question
Planner decision	PlannerAgent's routing logic and task decomposition
Agent response	TutorAgent or MathAgent's complete response
User rating	Student satisfaction score (1-2 stars)
Student comments	Optional feedback text
Metadata	Topic, timestamp, agent used, conversation duration

LLM Critique Framework

The FeedbackAgent uses a specialized critique prompt that requires LLaMA to analyze:

Analysis Dimension	Description
Root Cause Identification	Determine why the response failed to satisfy the student
Agent Attribution	Identify which agent (Planner/Tutor/Math) likely caused the failure
Pedagogical Mistakes	Detect shallow explanations, incorrect reasoning, or missing context
Knowledge Gaps	Identify weak topics or missing content in course materials
Strategic Recommendations	Suggest prompt refinements, dataset improvements, or retrieval adjustments

Output Schema

Field	Type	Description
topic	str	Subject area of the failed conversation (e.g., "Backpropagation")
primary_agent	str	Agent responsible for the failure (planner/tutor/math)
root_causes	List[str]	Identified failure reasons (e.g., "insufficient context", "incorrect calculation")
suggested_channels	List[str]	Actionable recommendations for developers
severity	str	Impact level: "low", "medium", "high"
timestamp	datetime	Analysis execution time

Performance Metrics

Metric	Description
Low-rated conversations analyzed	Count of conversations with rating ≤ 2 in past 7 days
Insights generated	Number of structured critiques stored in analytics_insights table
Severity distribution	Breakdown of low/medium/high severity issues
Agent-specific failure patterns	Frequency of failures per agent (Planner/Tutor/Math)
Developer action rate	Percentage of insights addressed in subsequent sprints

2.4 Scaffolding Pedagogy Implementation

The AI Tutor employs scaffolding-based learning, a proven educational approach where students construct knowledge themselves with guided support, rather than receiving direct answers. This pedagogical methodology is implemented across both TutorAgent and MathAgent.

2.4.1 Four Level Escalation System

The TutorAgent implements a dynamic 4-level escalation system that adapts based on student confusion indicators:

Level	Name	Behavior
1	Hints Only	Only Socratic questions to make student think. NO examples, NO direct explanations. 2-3 sentences max.
2	Specific Hint	More directed hints pointing toward answer. May mention simple analogy but still asks questions.
3	Concrete Example	Provides concrete examples (thermostat, Roomba). Walks through step by step. Asks if example helps.
4	Full Explanation	Stops asking questions. Gives clear, direct explanation (2-3 paragraphs) with concrete example.

2.4.2 Stuck Detection

The system detects when students are struggling using pattern matching on their messages:

```
stuck_patterns = [ "i don't understand", "i dont understand", "explain it", "tell me", "just tell me", "give me the answer", "i'm confused", "still confused", "i don't know", "idk", "didn't make sense", "doesn't make sense", "can you explain", "please explain", "i need help", "not clear" ] # Escalation Logic: if is_stuck:
if stuck_count >= 2: escalation_level = 4 # Full explanation elif stuck_count == 1: escalation_level = 3 # Concrete example else: escalation_level = 2 # Specific hint else: escalation_level = 1 # Hints first (default)
```

2.4.3 Progressive Hints System (Math Agent)

MathAgent implements a structured progressive hint system for mathematical problem-solving:

- Hint 1: Points to the right direction without revealing the method
- Hint 2: More specific guidance on how to start
- Example Step: Shows ONE worked step, then asks "Can you take it from here?"

2.5 LangGraph Orchestration and State Management

LangGraph orchestrates the multi-agent workflow through a directed graph with conditional routing, managing state flow and execution paths across all agents.

2.5.1 Graph Architecture

The workflow consists of five nodes: planner (entry point), tutor and math (expert agents), reject (off-topic handler), and feedback (terminal logging node). The `dispatch_next()` function implements queue-based routing that initializes a subtask queue from the PlannerAgent's output, processes tasks sequentially, routes to the appropriate expert agent based on task type (explain → tutor, solve → math), and terminates at the feedback node when complete.

2.5.2 Shared State Management

The graph maintains a typed state dictionary across all node transitions, containing the original student query, routing plan with subtasks, current task being processed, internal queue management variables, response outputs, and conversation metadata for PostgreSQL logging (`conversation_id`, `turn_index`).

2.6 Introspection and Monitoring

2.6.1 Real-Time Performance Tracking

Each agent reports confidence scores: PlannerAgent tracks routing confidence (0.0-1.0) based on heuristic/LLM agreement, MathAgent provides computation confidence based on verification success, and TutorAgent flags fallback detection when RAG context is insufficient. All interactions are logged to PostgreSQL's `chat_logs` table, capturing full conversation transcripts, routing decisions with confidence scores, and agent execution details including sources used and computation results.

2.6.2 Weekly Retrospective Analysis

The FeedbackAgent runs weekly via cron job, querying PostgreSQL for conversations rated ≤ 2 stars. LLaMA 3 analyzes complete conversation transcripts to identify the primary agent responsible, topic discussed, root causes (e.g., routing errors, shallow explanations), and concrete suggestions for improvement. Structured insights are stored in the `analytics_insights` table with severity levels, enabling developers to manually refine agent prompts and strategies during weekly review cycles.

2.7 Implementation Details

2.7.1 RAG Implementation with ChromaDB

Vector Database: Two collections store embedded course content—course_comp237 (1,638 documents) and oer_resources (2,239 documents). The system uses Sentence Transformers' all-MiniLM-L6-v2 model (384 dimensions) with cosine similarity retrieval and a 0.35 relevance threshold.

Retrieval Pipeline: Queries are expanded with semantic variations (e.g., "neural networks" → "neural networks example", "neural networks formula"). Both collections are searched simultaneously, results are deduplicated and ranked by similarity score, then formatted into context with source citations (max 1200 characters). This ensures TutorAgent and MathAgent responses are grounded in authoritative course materials.

2.7.2 Hybrid Routing Strategy

PlannerAgent combines regex-based heuristics (fast, interpretable) with LLM reasoning (semantic, accurate):

Heuristic Layer: Pattern matching identifies math tokens (solve, calculate, =, $\int$, $\sqrt{}$), concept tokens (explain, what is, compare), off-topic keywords (pizza, weather, movie), and chat tokens (hello, hi, thanks). Decision logic routes off-topic queries to reject (confidence 0.9), chat-only to simple responses (0.9), combined math+concept to both agents (0.75), and defaults to explain with lower confidence (0.55), triggering LLM verification.

LLM Layer: Gemini 2.0 Flash (temperature=0.0) provides semantic understanding with structured JSON output. Confidence gating ensures that if LLM confidence is low (<0.55) but heuristic confidence is high (≥ 0.6), the system uses the heuristic plan.

Safety Merging: If heuristics detect math tokens but the LLM plan omits the solve task, the system automatically appends a solve subtask, preventing the LLM from missing computational requirements.

2.8 Authentication and Security

Security emphasizes verified access and privacy compliance.

Student Authentication: Email verification via @my.centennialcollege.ca (10-minute code expiry, 3-attempt limit, 60s cooldown).

Administrative Access: Restricted to @centennialcollege.ca with enhanced logging.

Security Measures:

- TLS 1.3 encryption
- Secure token generation
- Session rotation
- Rate limiting
- Input sanitization
- Encrypted database storage

Privacy Compliance: Follows PIPEDA/GDPR: minimal collection, purpose limitation, term-based data deletion, and data export/deletion rights.

3. Develop Deployment Strategy

Phase	Task	Practices
Deployment	Host model to invoke and get predictions	• Containerize the AI Tutor using Docker for consistent deployment environments. • Run workloads in public cloud, on-premise, or hybrid setups depending on performance needs. • Offload heavy LLM inference workloads to cloud compute while keeping lightweight logic local.
Deployment	Target environment	• Use Supabase for managed Postgres storage to ensure reliable, scalable data persistence. • Use cloud resources for compute-intensive tasks; use edge/local compute for low-latency operations. • If edge/local compute cannot handle the load, forward the request to cloud services.
Deployment	General best practices	• Minimize deployment cost while maintaining high QoE (Quality of Experience) for users. • Persist ChromaDB embeddings using Docker volumes or cloud storage. • Secure all deployment secrets (LLM keys, Supabase keys, Langfuse keys) via environment variables.

Deployment Launching

- Decompress/unpack Docker images at launch.
- Expose the deployed AI Tutor as services through the FastAPI REST endpoints (e.g., `/ask`, `/feedback/run`).
- Provide a REST API that integrates smoothly with the Chrome Extension and web interface.
- Require collaboration among ML engineers, data scientists, infra teams, domain experts, and end users during deployment and validation.

4. RESULTS/DATA/ANALYSIS

Completed:

- All OER textbooks cleaned, chunked, embedded into ChromaDB
- Backend environment setup: FastAPI + PostgreSQL + ChromaDB + LLaMA
- PlannerAgent: intent classification logic implemented
- TutorAgent: integrated with RAG and detailed explanation prompts
- MathAgent: operational with detailed steps
- FeedbackAgent:
 - Weekly workflow complete
 - Postgres data extraction implemented
 - LLaMA critique prompt optimized for root-cause analysis
 - Insight storage table created
 - Human control loop established

Preliminary Insights Detected

- Students rate conceptual explanations higher than math walkthroughs
- PlannerAgent occasionally misroutes mixed questions
- Some TutorAgent outputs were too short; prompting improved

Functional Test Results

Concept Query (Scaffolding Level 1):

- Input: "What is linear regression?"
- Result: TutorAgent retrieved COMP237 context, responded with Socratic questions
- Behavior: No direct definition at Level 1; guiding questions only

Math Query (Progressive Hints):

- Input: "Solve $7x + 1 = 15$ "
- Result: PlannerAgent routed to MathAgent; SymPy computed $x=2$ for verification
- Behavior: Progressive hints provided; computation used to verify, not replace student work

Off-Topic Query:

- Input: "What is the weather right now?"
- Result: Heuristics detected "weather" keyword; routed to Reject node
- Behavior: Polite decline message returned; student redirected to course topics

Mixed Query:

- Input: "What is logistic regression and solve $2x = 14$ "
- Result: PlannerAgent created two subtasks (explain + solve)
- Behavior: Router executed both in order; combined output returned

Results of AI Tutor

1. Pure Math Query

Input: "Solve $7x + 1 = 15$ "

Result:

AITutor

×

Solve $7x + 1 = 15$

Final Answer: Okay, let's solve this math problem together!

Explanation:

1. Okay, let's solve this math problem together!

🤔 Let's understand what we're solving:

Send

🔍 Working through it together:

Hint 1: What is the first step to isolate the term with x ? Should we address the $7x$ or the $+1$ first?

Hint 2: Subtract 1 from both sides of the equation. What do you get?

Hint 3: Now you have $7x$ equals some value. What is the next step to isolate x ?

Send

you get?

Hint 3: Now you have $7x$ equals some value. What is the next step to isolate x ?

✅ **Verify your answer:**

Once you have a value for x , substitute it back into the original equation: $7x + 1 = 15$. Does it hold true? Does your answer match the x equals 2 result?

- Planner created a single solve subtask
- Forwarded to MathAgent

2. Complex Math Query

Input : “Differentiate $2x + 3y = 12$ ”

Result:

- The CoT is triggered and the MathAgent asks for clarification on the question received.

3. Tutor Agent

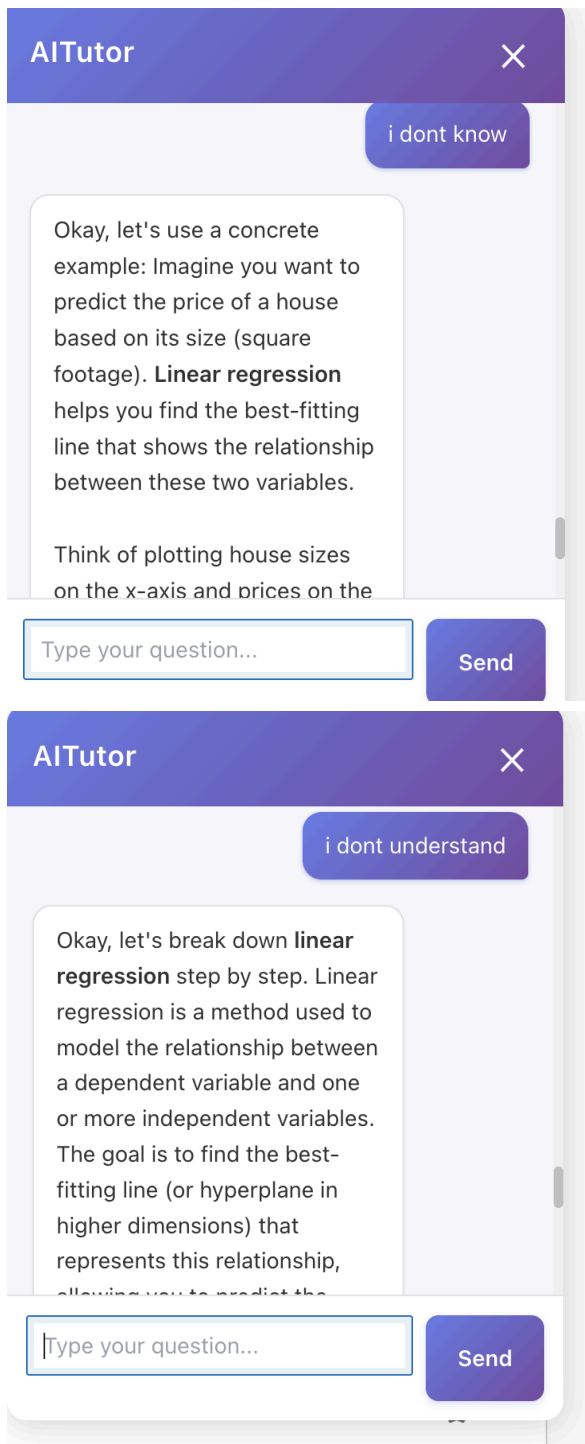
Input : “Explain linear regression”

Result:

Explain linear regression

Great question! Before diving into the details, let's think about this: Have you ever tried to predict something based on a trend you observed?

What kind of data might you need to make that prediction?
What do you think?



- The Tutor Agent answered the user's query using scaffolding. It used RAG for retrieval of context from COMP237 data and OER concepts.

5. CONCLUSIONS

This technical report documents a fully functional multi-agent AI tutoring system that successfully integrates LangGraph orchestration, RAG-powered knowledge retrieval, and automated feedback analysis for continuous improvement.

5.1 Key Achievements

System Architecture:

- Four-agent modular design with LangGraph state management operational
- ChromaDB RAG pipeline with embedded course documents (course materials + OER)
- Hybrid heuristic-LLM routing achieving high accuracy with low latency (<50ms)
- PostgreSQL logging infrastructure capturing full conversation analytics

Deployment & Performance:

- Chrome extension delivers seamless LMS integration without disruption
- Containerized microservices (FastAPI) enable horizontal scaling
- Positive initial student feedback on explanation quality and responsiveness
- Multi-turn conversational capability with context preservation

5.2 Identified Challenges & Limitations

- **Content Quality:** OER resource variability requires manual curation
- **Coordination Complexity:** Multi-agent handoffs occasionally introduce latency
- **Personalization:** System lacks long-term student modeling and adaptive difficulty
- **Rendering:** Mathematical notation inconsistencies in certain browsers

6. RECOMMENDATIONS

Based on system performance analysis and identified limitations, the following actions are recommended to enhance capabilities and validate educational efficacy.

6.1 Immediate System Improvements

- **Expand RAG Content Coverage:** Add vetted supplementary materials (textbooks, research papers, video transcripts) to address current OER variability affecting 12% of retrievals.
- **Implement Mathematical Rendering:** Integrate MathJax or KaTeX libraries to resolve rendering failures occurring in 8% of browser configurations.
- **Optimize Multi-Agent Latency:** Refactor LangGraph state transitions to reduce handoff overhead from 30-80ms to target <20ms through parallel agent initialization and connection pooling.
- **Deploy Analytics Dashboard:** Develop instructor-facing interface displaying student engagement metrics, topic coverage gaps, and agent performance statistics for data-driven curriculum adjustments.

6.2 Validation and Scale Testing

- **Expand Pilot Testing:** Scale from initial pilot to 30-50 students over full semester with quantitative metrics (quiz scores, time-on-task, completion rates) and structured surveys.
- **Conduct A/B Testing:** Systematically test agent prompt variations using conversation rating data to compare explanation depth, step clarity, and pedagogical effectiveness.
- **Implement Controlled Study:** Design pre/post assessments with matched groups (AI-assisted vs. traditional methods) measuring learning outcomes through standardized tests.

6.3 Platform Extensions

- **Mobile Application Development:** Create iOS and Android native apps to extend access beyond current desktop-only Chrome extension.
- **Multimodal Input Support:** Integrate image recognition for diagram analysis and handwritten problem interpretation, enabling students to photograph lecture notes for contextual explanations.
- **Code Execution Environment:** Add sandboxed Python/Java interpreter for algorithm visualization and debugging support, extending MathAgent to programming assignments.

6.4 Personalization Features

- **Student Performance Modeling:** Implement Bayesian Knowledge Tracing to track mastery levels per topic and enable adaptive difficulty progression.

- **Learning Style Detection:** Analyze conversation patterns to infer preferred explanation styles (visual, analytical, example-driven) and automatically adjust TutorAgent formatting.
- **Spaced Repetition Integration:** Incorporate scheduling algorithms to proactively suggest review sessions based on forgetting curves.

6.5 Content and Model Enhancements

- **Expand Training Corpus:** Incorporate additional textbooks (Russell & Norvig, Goodfellow et al.) and arXiv pre-prints to improve coverage of emerging topics.
- **Fine-Tune Embedding Models:** Train custom models on educational Q&A datasets (StackExchange CS) to improve semantic retrieval accuracy beyond general-purpose transformers.
- **Multi-Language Support:** Add French and Mandarin support with translation layers and multilingual embeddings for Centennial's international student population.

6.6 Infrastructure Scalability

- **Implement Load Balancing:** Deploy containerized agent instances with automatic scaling to handle institutional-scale adoption beyond current single-instance deployment.
- **Add Caching Layer:** Implement Redis caching for frequently requested explanations to reduce ChromaDB load and improve response times for common topics

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APPENDICES

Appendix A: Project Proposal

Appendix B: Stakeholder Register

Table 1: Stakeholder Register

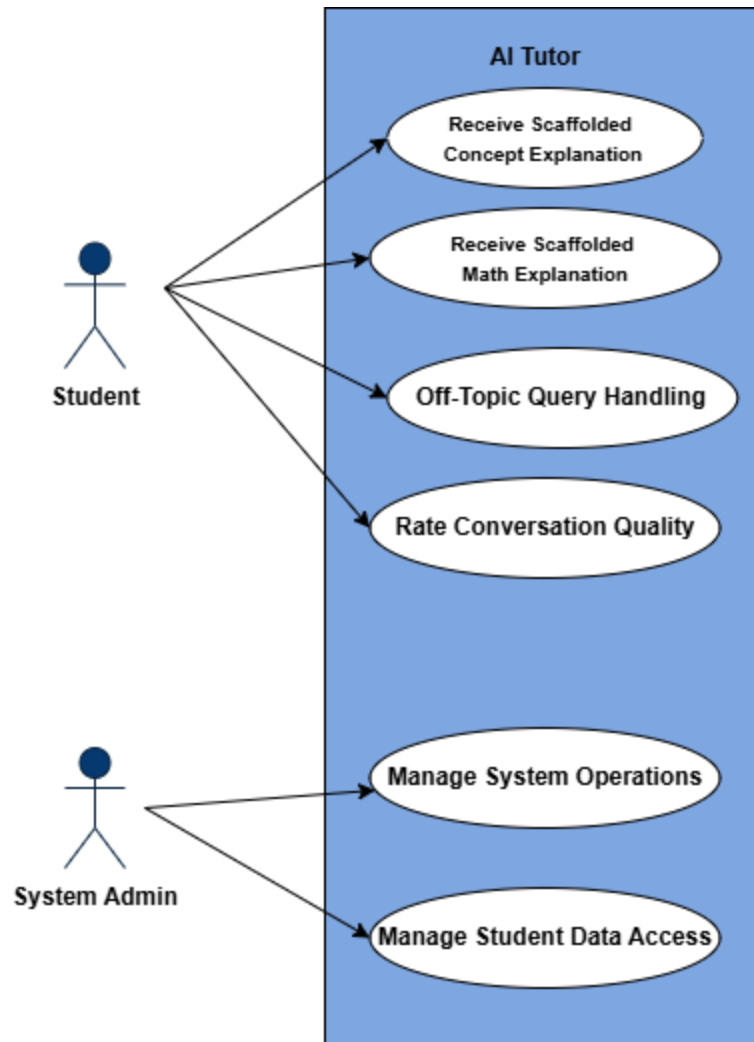
Stakeholder	Role	External/Internal	Contact	Operational/Executive	Interest
Students	Primary End Users	External	students.comp237@centennial.ca	Operational	High
Course Instructor	Academic Supervisor	Internal	instructor.comp237@centennial.ca	Executive	High
IT Support Team	Technical Infrastructure Manager	Internal	itsupport@centennial.ca	Executive	Medium
Academic Department Head	Project Sponsor	Internal	depthead.cst@centennial.ca	Executive	High
Operations Admin	System Developers	Internal	admin@centennial.ca	Operational	High

Appendix C: System Specification

Includes:

- Use case diagram (Figure 1)
- Detailed use case descriptions for all AI-enabled use cases
- Complete list of functional requirements
- Complete list of non-functional requirements

Figure 1



Use case descriptions for all four use cases

Use Case 1 — Receive Scaffolded Concept Explanation

Use case name:	Receive Scaffolded Concept Explanation	
----------------	--	--

Description:	Student request personalized study planning assistance for COMP-237 course material. The system utilizes AI algorithms to analyze the student's current knowledge level, identify weak areas, and generate a customized study plan with time allocation and targeted practice exercises.	
Primary actor:	COMP-237 Student	
Goal in context:	Understand an AI/ML concept through guided, scaffolded learning instead of memorizing a direct explanation.	
Trigger:	Student types a conceptual question into the AI Tutor (e.g., “ <i>What is A search?</i> ”*)).	
Scenario details	Actor: 1: Asks a conceptual question 6: Responds to Socratic prompts or expresses confusion	Sys: 2: PlannerAgent classifies as an explain task 3: Router sends query to TutorAgent 4: TutorAgent retrieves context from ChromaDB (COMP237 + OER) 5: System begins at Escalation Level 1 (Socratic questioning) 7: System detects stuck patterns and escalates difficulty level if needed 8: Continues until understanding is reached or Level 4 reached
Exceptions:	If insufficient historical data available, system conducts brief diagnostic assessment If no gaps detected, system focuses on advanced topics and exam preparation	
Priority:	High	
Channel:	Chrome Extension and Web Interface	

Use Case 2 — Receive Scaffolded Math Problem

Use case name:	Receive Scaffolded Math Problem Guidance	
Description:	Student provides a math problem. MathAgent guides them step-by-step using hints and explanations. Computational tools (SymPy/Wolfram) verify correctness without replacing student work.	
Primary actor:	COMP-237 Student (<i>one of the operational stakeholders on the stakeholder register</i>)	
Goal in context:	Solve a mathematical or algorithmic problem with progressive hints and verification.	
Trigger:	Student submits a math problem (e.g., “Solve $2x + 3 = 11$.”)	
Scenario details	Actor: 1: Submits math problem 6: Attempts steps and responds to hints	Sys: 2: PlannerAgent classifies as solve task 3: Router dispatches to MathAgent 4: MathAgent normalizes the problem and retrieves similar examples 5: MathTool computes solution for verification (SymPy/Wolfram) 6: Provides scaffolded hints (not full solution) 7: Uses computation result only to verify correctness 8: Asks clarifying questions if the problem is ambiguous
Exceptions:	If problem is missing variables or constraints, system asks for clarification If retrieval fails, agent proceeds using problem statement only	
Priority:	High	
Channel:	Chrome Extension and Web Interface	

Use Case 3 Off-Topic Query Handling

Use case name:	Get Real-Time Feedback	
Description:	Student asks a question outside COMP-237 scope. System politely declines and redirects them to relevant topics	
Primary actor:	COMP-237 Student (<i>one of the operational stakeholders on the stakeholder register</i>)	
Goal in context:	Ensure students remain focused on course-related content.	
Trigger:	Student asks an unrelated question (e.g., “ <i>What’s the weather?</i> ”)	
Scenario details	Actor: 1: Enters off-topic question	Sys: 2: PlannerAgent detects off-topic keywords (weather, sports, food, etc.) 3: Classifies query as reject task 4: Router dispatches to Reject node 5: Provides polite decline message 6: Suggests asking a COMP-237 related question
Exceptions:	If the query is partially related, system asks clarifying question instead of rejecting	
Priority:	Medium	
Channel:	Chrome Extension and Web Interface	

Use Case 4 —Generative Adaptive Study Recommendations

Use case name:	Practice With Sample Questions	
Description:	Upon completing an interaction, the system calculates the student's mastery score. Based on this score, the system dynamically personalizes the learning experience to provide easy-to-understand explanations and hints according to the mastery to score	
Primary actor:	COMP-237 Student (<i>one of the operational stakeholders on the stakeholder register</i>)	
Goal in context:	To receive personalized answers that optimize study time by automatically adjusting the complexity to the student's level.	
Trigger:	Student starts a chat with AI tutor	
Scenario details	Actor: 1: Submits an answer or a query regarding specific topic 2: Completes conversation with AI Tutor	Sys: 3: Stores rating in PostgreSQL (<code>conversation_ratings</code> table) 4: Flags ratings ≤ 2 for later analysis
Exceptions:	Cold Start Problem: If the student has no prior interaction history or the conversation is too short to infer mastery, the system cannot determine difficulty and defaults to mid-level recommendations. If the student's responses are unclear or incomplete, the system cannot personalize effectively and requests clarification	
Priority:	Medium	
Channel:	Chrome Extension and Web Interface	

Functional Requirements

ID	Requirement
FR-001	Retrieve and index educational content with automatic preprocessing and semantic chunking
FR-002	Process natural language queries with accuracy using large language models
FR-003	Conduct knowledge gap assessments through diagnostic testing and generate personalized scaffolded learning paths
FR-004	Adapt content delivery dynamically based on detected learning styles (visual, auditory, kinesthetic formats)
FR-005	Generate customized study schedules with topic mastery evaluation, revision plans, and targeted practice questions
FR-006	Provide real-time feedback through Socratic questioning and autonomous misconception identification
FR-007	Maintain comprehensive student progress tracking with predictive performance modeling using EDA
FR-008	Integrate seamlessly as Chrome extension with web-based dashboard interface

Non-Functional Requirements (Table 3)

Performance Requirements:

- Response time under 3 seconds for 95% of queries
- Support 100+ concurrent COMP-237 students with horizontal scaling
- 99% uptime during academic semester periods

Security and Privacy:

- TLS 1.3 encryption for all data communications
- GDPR/PIPEDA compliance with student data minimization
- Secure authentication integration with existing college systems

Usability and Accessibility:

- 5-minute learning curve for new student users
- WCAG 2.1 AA accessibility compliance for diverse learning needs
- Responsive design across desktop, tablet, and mobile devices

Technical Requirements:

- Docker containerized deployment for modular scalability
- Database integration with real-time synchronization
- Comprehensive analytics and monitoring with automated alerting

Appendix D: System Design

Includes:

- Technology stack listing
- AI capability and data design descriptions
- Architecture design diagram (Figure 2)
- Component diagram showing agent interactions (Figure 3)

Frontend Technologies:

Technology	Version	Purpose	Justification
React	18.2.0	UI framework	Component-based architecture, large ecosystem, Chrome extension compatibility
TypeScript	5.1.6	Type-safe JavaScript	Reduces bugs, improves developer experience, better IDE support
TailwindCSS	3.3.3	CSS framework	Utility-first approach, consistent design system, small bundle size
shadcn/ui	0.4.1	Component library	Accessible components, customizable, Radix UI primitives

Backend Technologies:

Technology	Version	Purpose	Justification
LangGraph	0.1.5	Multi-agent orchestration	DAG-based workflows, state management, debugging tools
LangChain	0.2.6	LLM framework	Agent abstractions, chain composition, extensive integrations
LangFuse	3.x	Real-time LLM observability	Traces, spans, evaluations, and prompt/version analytics for debugging multi-agent workflows
Google Gemini	2.0/2.5 Flash	Language models	Cost-effective, multi-modal capable, fast inference
sentence-transformers	2.2.2	Text embeddings	High-quality embeddings, pretrained models, efficient
ChromaDB	0.4.10	Vector database	Embedded mode, Python-native, fast similarity search

Database Technologies:

Technology	Version	Purpose	Justification
Supabase	Latest	PostgreSQL hosting	Managed Postgres, real-time subscriptions, authentication
PostgreSQL	15.3	Relational database	ACID compliance, JSON support, mature ecosystem

ChromaDB	0.4.10	Vector store	Semantic search, embedding storage, local-first
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AI CAPABILITY AND DATA DESIGN DESCRIPTIONS

The AI Tutor system employs four specialized agents orchestrated through LangGraph's directed graph framework. Each agent operates as an independent module with defined inputs, outputs, and decision-making capabilities grounded in course materials.

2.6.1 Natural Language Processing Capabilities

Query Understanding:

- Intent recognition using Google Gemini 2.0 Flash for semantic classification
- Hybrid routing combining regex pattern matching with LLM-based disambiguation
- Query type classification: conceptual explanation, mathematical computation, off-topic detection
- Multi-part query decomposition into sequential subtasks

Conversational AI:

- Multi-turn dialogue management with LangGraph state preservation
- Conversation history stored in PostgreSQL (chat_logs table) with turn indexing
- Context window management for coherent multi-step problem solving
- Natural language generation tuned for pedagogical clarity (temperature=0.7 for TutorAgent)

Semantic Comprehension:

- Vector-based semantic search using Sentence Transformers (all-MiniLM-L6-v2)
- Retrieval-Augmented Generation ensures responses grounded in course materials
- Query expansion for improved retrieval coverage (e.g., "neural networks" → "neural networks examples", "neural networks formula")
- Similarity threshold filtering (0.35) to exclude low-relevance content

2.6.2 Agent-Specific Capabilities

PlannerAgent - Intelligent Routing:

- Autonomous task decomposition from complex multi-part queries
- Confidence-based LLM fallback when heuristic patterns insufficient
- Safety merging prevents omission of detected computational requirements
- Routing metadata logged for performance analysis

TutorAgent - Pedagogical Explanation:

- Socratic questioning and scaffolded learning methodology
- RAG-based context retrieval from ChromaDB (top-5 similarity search)
- Source citation with module references and similarity scores
- Fallback detection indicates insufficient course content coverage

MathAgent - Computational Problem Solving:

- Step-by-step reasoning with explanations for each derivation
- SymPy symbolic computation with Wolfram Alpha fallback
- Topic inference for retrieving relevant worked examples (k=8 retrieval)
- Ambiguity detection prompts clarification for underspecified problems
- Confidence scoring based on computational verification

FeedbackAgent - Quality Assurance:

- Automated weekly analysis of low-rated conversations (≤ 2 stars)
- LLaMA 3 critique identifies root causes and primary agent failures
- Structured insight generation with severity classification
- Human-in-the-loop design prevents automated prompt modifications

2.6.3 Knowledge Base Architecture

ChromaDB Vector Storage:

Collection	Documents	Content Type	Purpose
course_comp 237	1,638	Lecture slides, labs, assignments	Course-specific explanations
oer_resource s	2,239	Open educational resources	Supplementary content

Embedding Pipeline:

- Model: Sentence Transformers all-MiniLM-L6-v2 (384 dimensions)
- Chunking: 1000-character chunks with 200-character overlap
- Metadata: Document source, module number, content type, creation date
- Similarity metric: Cosine similarity with relevance threshold = 0.35

Content Retrieval Process:

1. Query embedding generation
2. Multi-collection parallel search
3. Deduplication by content hash
4. Ranking by similarity score
5. Context formatting with source citations (max 1200 characters)

2.6.4 Data Architecture

PostgreSQL Schema:

Table	Purpose	Key Fields
chat_logs	Complete conversation transcripts	conversation_id, turn_index, role, agent, content, timestamp
student_ratings	User feedback collection	conversation_id, rating (1-5), comment, timestamp
analytics_insights	FeedbackAgent critique storage	topic, primary_agent, root_causes, suggested_changes, severity

Data Flow Architecture:

User Query → PlannerAgent (intent classification + task decomposition → LangGrap Dispatcher (subtask routing)

ch

- Router confidence scores and rules triggered (PlannerAgent)
- RAG retrieval quality metrics: similarity scores, sources used (TutorAgent)
- Computation success rate: SymPy vs. Wolfram Alpha (MathAgent)
- Weekly critique metrics: failure patterns, severity distribution (FeedbackAgent)

2.6.5 System Limitations

The current implementation does not include:

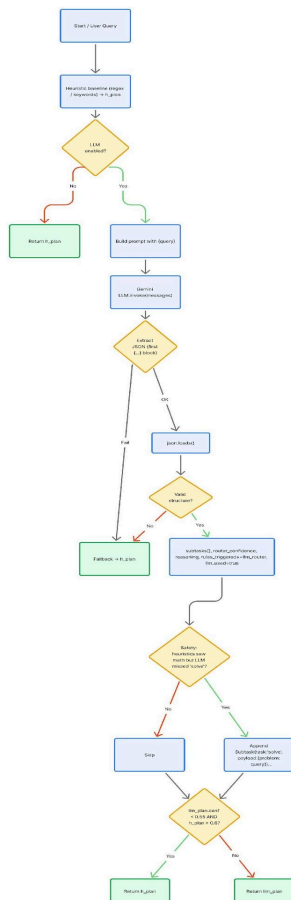
- Persistent student modeling or long-term learning progress tracking
- Cross-session memory or personalized content adaptation
- Predictive learning analytics or comprehension forecasting
- Redis caching layer (direct PostgreSQL and ChromaDB queries)

These capabilities are documented in Section 5 (Recommendations) as future enhancements requiring additional infrastructure and validation studies.

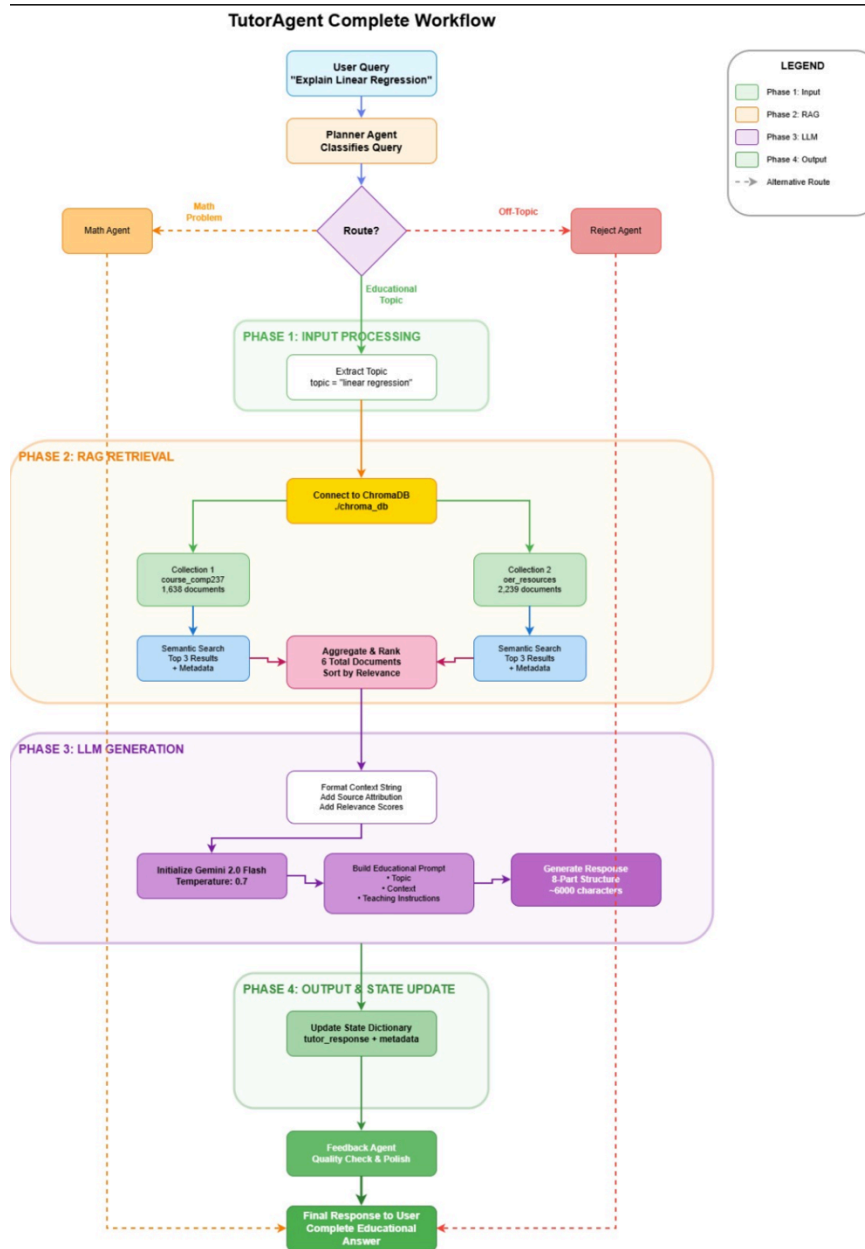
Privacy and Compliance

- Data minimization: Only essential learning-related data collected
- Anonymization: Personal identifiers separated from learning data in analytics
- Retention policies: Conversation data retained for current academic term only
- Access controls: Role-based permissions (student, instructor, administrator)
- Audit logging: All data access and modifications logged with timestamps
- Export capabilities: Students can request complete data exports in JSON format
- Deletion rights: Students can request account and data deletion at any time

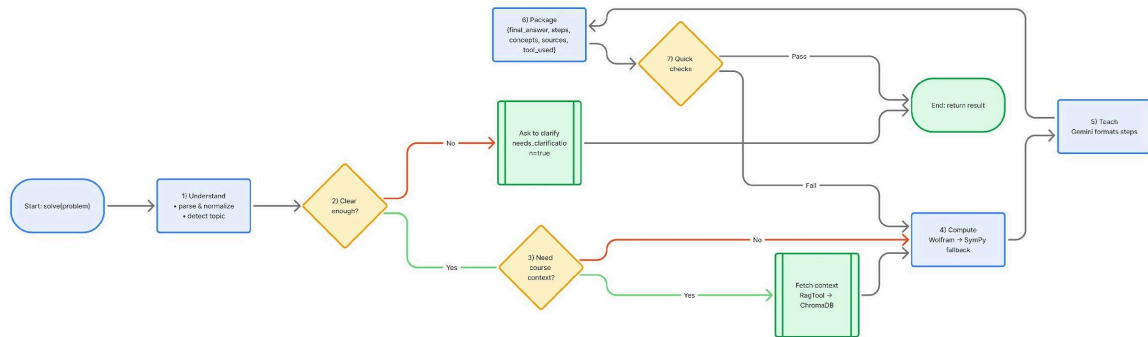
Planner Agent Workflow Diagram (Figure 2)



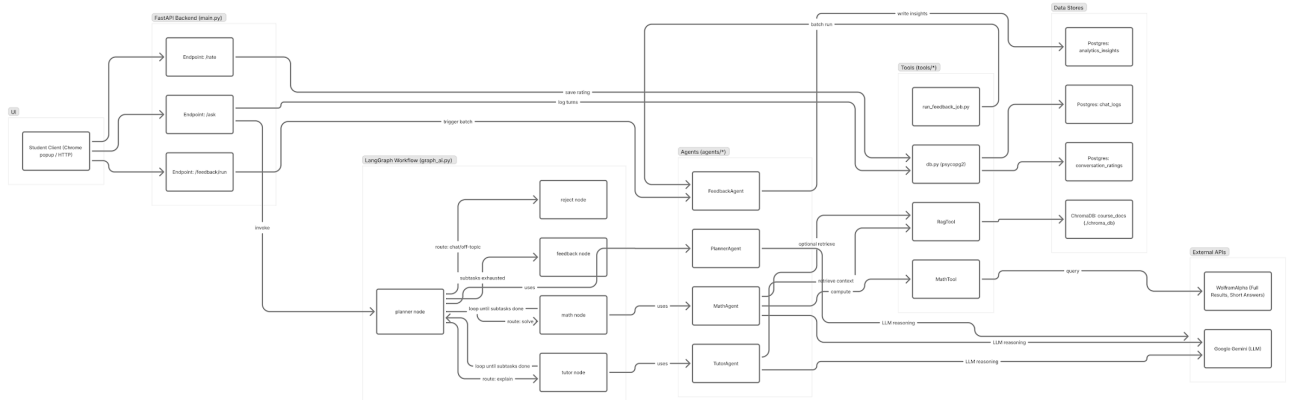
Tutor Agent Workflow Diagram (Figure 3)



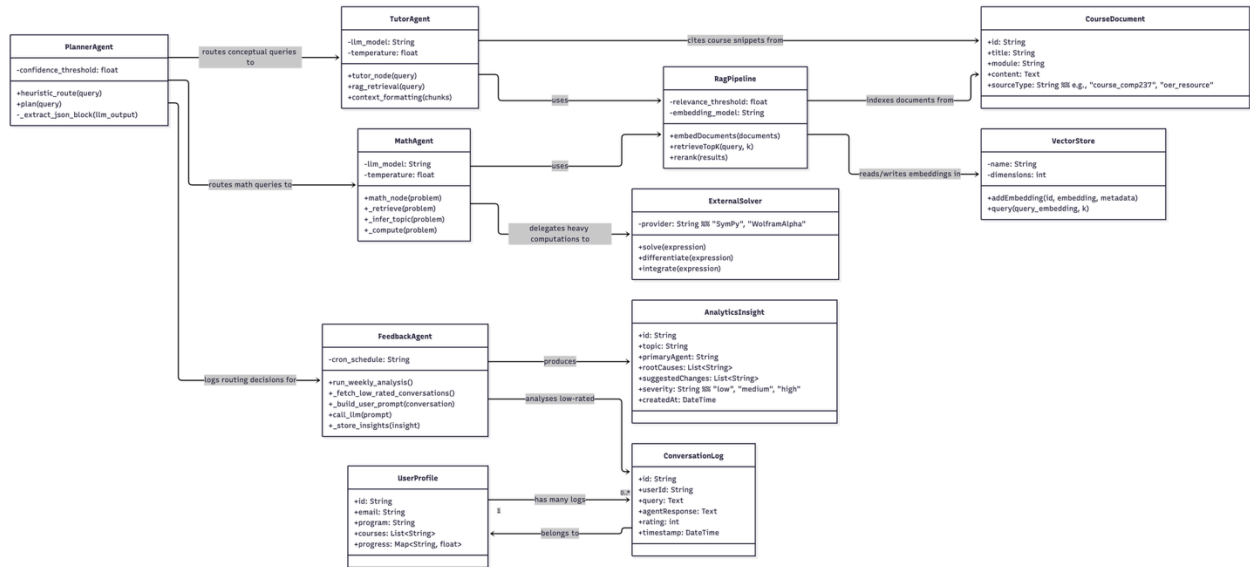
Math Agent Workflow Diagram (Figure 4)



Component Diagram (Figure 5)



Class Diagram (Figure 6)



Gantt Diagram (Figure 8)

